

Домашняя работа по ТРКП  
 Выяснить элементарным способом

№ 1.23

№ 1.32  $\operatorname{Im} \frac{z-z_1}{z-z_2} = 0 \Rightarrow z_2 \neq z$

$\operatorname{Re} \frac{z-z_1}{z-z_2} = 0 \Rightarrow z_2 \neq z$

$$\operatorname{Im} \frac{x+yi-z_1}{x+yi-z_2} = \operatorname{Im} \frac{(x+yi-z_1)(x-yi-z_2)}{(x+yi-z_2)(x-yi-z_2)} =$$

$$= \operatorname{Im} \frac{x^2 - z_2x + y^2 - z_1x + z_2z_1 + (-z_2y + z_1y)i}{x^2 - 2z_2x + z_2^2 + y^2} =$$

$$= \frac{(-z_2y + z_1y)i}{x^2 - 2z_2x + z_2^2 + y^2} = \frac{y(z_1 - z_2)}{(x - z_2)^2 + y^2} = 0$$

наилучший результат -  
 прямой, проходящий через  
 $z_1$  и  $z_2$

$y(z_1 - z_2) = 0, z_2 \neq z$

$$\operatorname{Re} \frac{z-z_1}{z-z_2} = \frac{x^2 - 2z_2x + y^2 - z_1x + z_1z_2}{(x - z_2)^2 + y^2} = 0$$

$$x^2 - (z_1 + z_2)x + \frac{(z_1 + z_2)^2}{4} - \frac{(z_1 + z_2)^2}{4} + z_1z_2 + y^2 = 0$$

$$\left(x - \frac{z_1 + z_2}{2}\right)^2 + y^2 = \frac{(z_1 + z_2)^2}{4} - z_1z_2$$

результат - окружность, из которой вытекает  $z_2$

$$R = \sqrt{\frac{(z_1 + z_2)^2}{4} - z_1z_2} = \frac{z_1 - z_2}{2} \Rightarrow D = 2R = z_1 - z_2$$

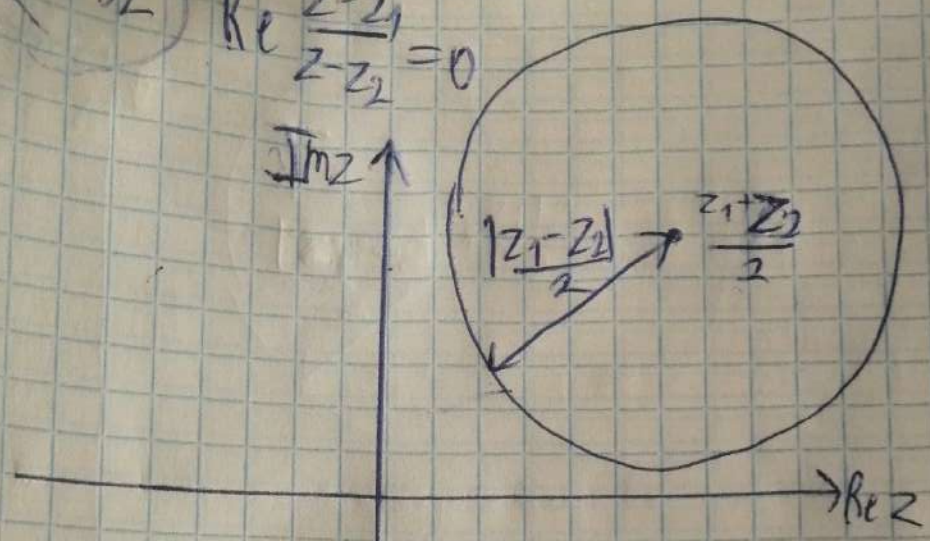
(м-с,  
 планетарная окружность с центром  
 в  $z_1$  и  $z_2$ )



N1.32

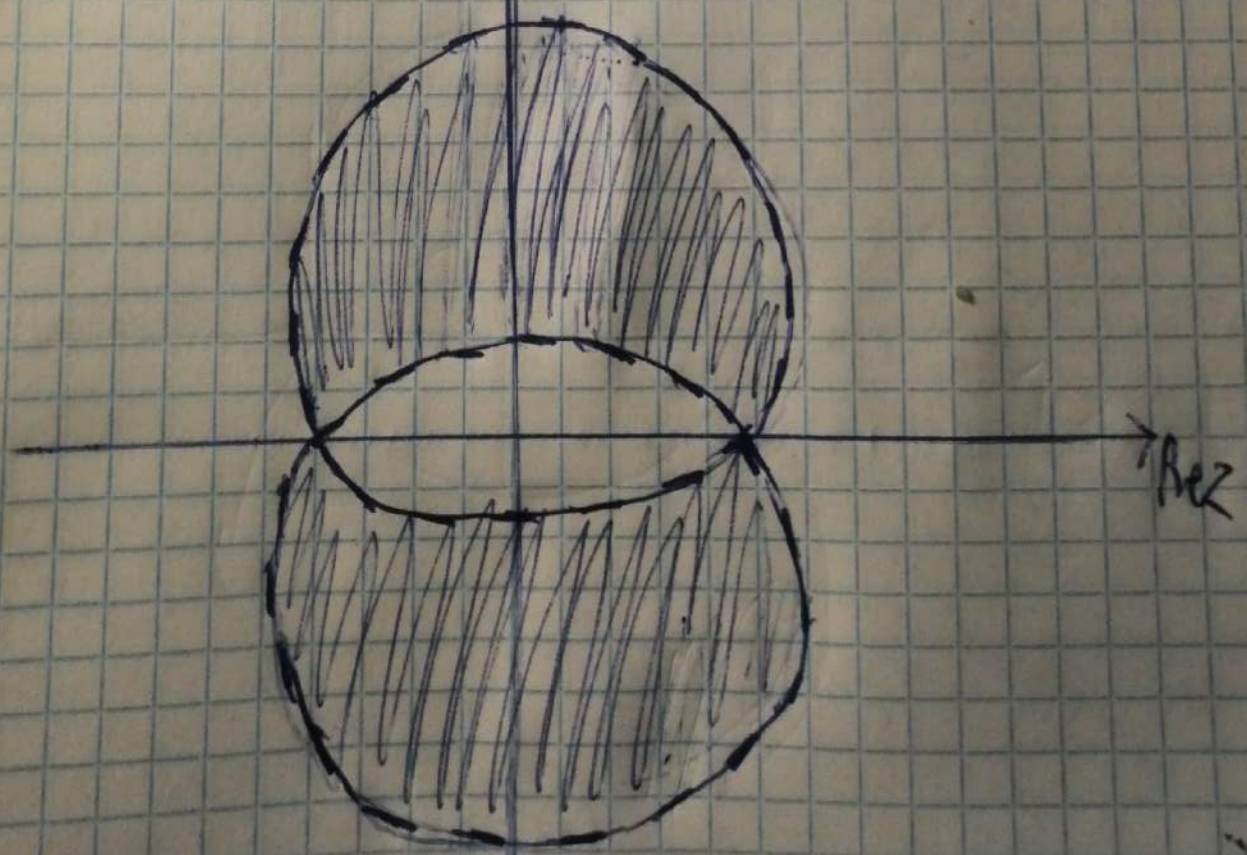
N1.32

$$\operatorname{Re} \frac{z-z_1}{z-z_2} = 0$$



N1.33

$\operatorname{Im} z$





N1.33

$$|2z| > |1+z^2|$$

$$z = x+yi$$

$$|2z| > |1+(x+yi)^2|$$

$$|2z| > |1+x^2+2xyi-y^2|$$

$$|1+z^2| - |2z| < 0$$

$$|2(x+yi)| > |1+x^2+2xyi-y^2|$$

$$|2(x+yi)| > |1+(x-yi)^2|$$

$$2\sqrt{x^2+y^2} > |1+(x-yi)^2|$$

$$2\sqrt{x^2+y^2} > |1+x^2-y^2+2xyi|$$

$$2\sqrt{x^2+y^2} > \sqrt{(1+x^2-y^2)^2+4x^2y^2}$$

если  $2z \leq 0$ , то  $1+z^2+2z < 0$   $(z+1)^2 < 0$

если  $2z > 0$ , то  $1+z^2-2z < 0$   $(z-1)^2 < 0$

Перепишем соотношения в комплексной форме.

$$z^2+2zi+1 < 0 \Rightarrow z^2+2zi-1 < -2 \quad (z+i)^2 \geq 2$$

$$z^2-2zi+1 < 0 \Rightarrow z^2-2zi-1 < -2 \quad (z-i)^2 \geq 2$$

Если квадрат отрицателен, то модуль комплексной функции  $|z+i| = \sqrt{2}$  и  $|z-i| = \sqrt{2}$

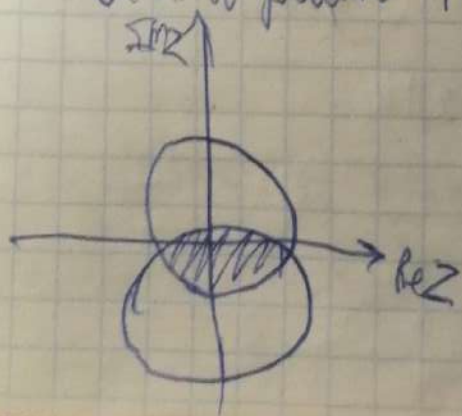
$$|z+i| = |x+yi+i| = |x+(y+1)i| = \sqrt{x^2+(y+1)^2} = \sqrt{2}$$

$$|z-i| = |x+yi-i| = |x+(y-1)i| = \sqrt{x^2+(y-1)^2} = \sqrt{2}$$

В действительной форме  $x^2+(y+1)^2=2$  и  $x^2+(y-1)^2=2$

П.к. по условию  $|2z| > |1+z^2| \Rightarrow$  указанная область

- это внутренность пересечения двух окружностей радиуса  $\sqrt{2}$  с центрами в  $i$  и  $-i$ .





Найти все значения многозначных элементов.  
 $e^{ip} = \cos p + i \sin p$

N 1.74

$$1) (x+iy)^{(u+vi)} = z^n = |z|^n (\cos np + i \sin np)$$

$$= e^{(u+vi) \ln(x+iy)} = e^{(u+vi) [\ln \sqrt{x^2+y^2} + i \operatorname{arctg} \frac{y}{x} + 2\pi k]} = e^{\sqrt{2} \ln 1 + i \cdot 2\pi k} = e^{\sqrt{2} i 2\pi k} = (\cos(2\sqrt{2}\pi k) + i \sin(2\sqrt{2}\pi k))$$

ответ:  $1^{\sqrt{2}} = \cos(2\sqrt{2}\pi k) + i \sin(2\sqrt{2}\pi k)$

$$2) (-2)^{\sqrt{2}} = e^{\sqrt{2} \ln(-2)} = e^{\sqrt{2} \ln 2 + i \sqrt{2} \pi (2k+1)} = 2^{\sqrt{2}} (\cos(\sqrt{2}\pi(2k+1)) + i \sin(\sqrt{2}\pi(2k+1)))$$

ответ:  $(-2)^{\sqrt{2}} = 2^{\sqrt{2}} (\cos(\sqrt{2}\pi(2k+1)) + i \sin(\sqrt{2}\pi(2k+1)))$

$$3) 2^i = e^{i \ln 2} = \cos \ln 2 + i \sin \ln 2$$

ответ:  $2^i = \cos \ln 2 + i \sin \ln 2$

$$4) 1^{-i} = e^{-i \ln 1} = e^{2\pi k}$$

ответ:  $1^{-i} = e^{2\pi k}$



$$5) i^i = e^{i[\ln 1 + i(2\pi k + \arctan \frac{1}{0})]} = e^{i[i(2\pi k + \arctan \infty)]} = e^{-[2\pi k + \frac{\pi}{2}]} = e^{(2k - \frac{1}{2})\pi}$$

Answer:  $i^i = e^{(2k - \frac{1}{2})\pi}$

$$6) \left(\frac{1-i}{\sqrt{2}}\right)^{1+i} = \left(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right)^{1+i} = e^{(1+i)[\ln 1 + i(\arctan(-1) + 2\pi k)]}$$

$$= e^{(1+i)(-\frac{\pi}{4} + 2\pi k)i} = e^{(-\frac{\pi}{4} + 2\pi k)i + (-\frac{\pi}{4} - 2\pi k)} = e^{-\frac{\pi}{4} - 2\pi k} e^{(\frac{\pi}{4} + 2\pi k)i}$$

$$\left(\frac{1-i}{\sqrt{2}}\right)^{1+i} = \left(\frac{1-i}{\sqrt{2}}\right)^i e^{i[\ln 1 + i(-\frac{\pi}{4} + 2\pi k)]} = \left(\frac{1-i}{\sqrt{2}}\right)^i e^{(2k + \frac{1}{4})\pi}$$

Answer:  $\left(\frac{1-i}{\sqrt{2}}\right)^{1+i} = \left(\frac{1-i}{\sqrt{2}}\right)^i e^{(2k + \frac{1}{4})\pi}$

$$7) (3-4i)^{1+i} = e^{(1+i)[\ln 5 + i(\arctan \frac{-4}{3} + 2\pi k)]} =$$

$$= e^{(\ln 5 + i(\arctan \frac{-4}{3}) + 2\pi k) + \ln 5i + i(\arctan \frac{4}{3} + 2\pi k)}$$

$$= 5e^{\arctan \frac{4}{3} + 2\pi k} e^{(\ln 5 - \arctan \frac{4}{3} + 2\pi k)i} = 5e^{\arctan \frac{4}{3} + 2\pi k} (\cos(\ln 5 - \arctan \frac{4}{3}) + i \sin(\ln 5 - \arctan \frac{4}{3}))$$

Answer:  $(3-4i)^{1+i} = 5e^{\arctan \frac{4}{3} + 2\pi k} (\cos(\ln 5 - \arctan \frac{4}{3}) + i \sin(\ln 5 - \arctan \frac{4}{3}))$

$$8) (-3+4i)^{1+i} = e^{(1+i)[\ln 5 + i(\arctan \frac{4}{-3} + 2\pi(k+1))]} = e^{(\ln 5 + \arctan \frac{4}{-3} - 2\pi(k+1))}$$

$$= e^{(\ln 5 - \arctan \frac{4}{3} - 2\pi(k+1))} = e^{-(\ln 5 + \arctan \frac{4}{3} + 2\pi(k+1))} (\cos(\ln 5 - \arctan \frac{4}{3}) + i \sin(\ln 5 - \arctan \frac{4}{3}))$$

Answer:  $(-3+4i)^{1+i} = -5e^{\arctan \frac{4}{3} + (2k+1)\pi} (\cos(\ln 5 - \arctan \frac{4}{3}) + i \sin(\ln 5 - \arctan \frac{4}{3}))$



№1.122 Найти преобразование  $w = z^2 + z$  точки  $z = c$  на плоскости  $z$  с помощью функции

$$(1) w = z^2 + z$$

$$z = x + yi$$

$$w = (x + yi)^2 + x + yi = x^2 + 2xyi - y^2 + x + yi = (x^2 - y^2 + x) + (2xy + y)i$$

$$w = u + iv$$

$$\text{при } x = c$$

$$u = x^2 - y^2 + x = c^2 - y^2 + c$$

$$v = 2xy + y = 2cy + y$$

$$v = y(2c + 1)$$

$$y = \frac{v}{2c + 1}$$

$$u = c^2 - \frac{v^2}{(2c + 1)^2} + c \quad (1)$$

$$\text{при } y = c$$

$$u = x^2 - y^2 + x = x^2 - c^2 + x = \frac{(v - c)^2}{4c^2} - c^2 + \frac{v - c}{2c} \quad (2)$$

$$v = y(2x + 1) = c(2x + 1) = 2xc + c \Rightarrow x = \frac{v - c}{2c}$$



$$(1) \quad u = \frac{c^2(2c+1)^2 - v^2 + c(2c+1)^2}{(2c+1)^2}$$

$$v^2 = c^2(2c+1)^2 + c(2c+1)^2 - u(2c+1)^2 =$$

$$= (2c+1)^2 (c^2 + c - u)$$

$$\text{при } a = c + \frac{1}{2} \Rightarrow c = a - \frac{1}{2}$$

$$(2(c + \frac{1}{2}))^2 ((a - \frac{1}{2})^2 + a - \frac{1}{2} - u) = v^2$$

$$4a^2(a^2 - u - \frac{1}{4}) = v^2$$

$$-4a^2(u + \frac{1}{4} - a^2) = v^2 \quad \text{— семейство парабол с вершиной в точке } u = -\frac{1}{4}$$

$$\text{при } a = 0 \quad v^2 = 0 \quad \text{— лр, исходящий из точки } u = -\frac{1}{4} \text{ по ординате } R \text{ ось } (a=0)$$

$$-4a^2(u + \frac{1}{4} - a^2) = 0$$

$$(2) \quad u = \frac{(v-c)^2}{4c^2} - c^2 + \frac{v-c}{2c}$$

$$u = \frac{(v-c)^2 - 4c^4 + (v-c)2c}{4c^2}$$

$$4c^2 u = (v^2 - 2cv + c^2) - 4c^4 + 2cv - 2c^2$$

$$4c^2 u = v^2 - 4c^4 - 2c^2$$

$$v^2 = 4c^2 u + 4c^4 + 2c^2 = 4c^2(u + c^2 + \frac{1}{2}) \quad \text{— семейство сопряженных парабол, при } c=0 \Rightarrow \text{лр, исходящий из точки } u = -\frac{1}{2} \text{ по } R \text{ ось в полне. черпаку.}$$



$$2) W = \coth z = \frac{1}{\tanh z} = \frac{\cosh z}{\sinh z} = \frac{e^z + e^{-z}}{e^z - e^{-z}} = \frac{e^{2z} + 1}{e^{2z} - 1}$$

$$W = \coth(x+yi)$$

Из формулы Эйлера:  $e^{ip} = \cos p + i \sin p$

$$e^{2z} = e^{2(x+yi)} = e^{2x} \cdot e^{2yi} = e^{2x} (\cos 2y + i \sin 2y)$$

$$W = \frac{e^{2x} (\cos 2y + i \sin 2y) + 1}{e^{2x} (\cos 2y + i \sin 2y) - 1} = \frac{(e^{2x} (\cos 2y + i \sin 2y) + 1)(e^{2x} (\cos 2y - i \sin 2y))}{(e^{2x} (\cos 2y + i \sin 2y) - 1)(e^{2x} (\cos 2y - i \sin 2y))}$$

$$= \frac{e^{4x} - 2e^{2x} i \sin 2y - 1}{e^{4x} - 2e^{2x} \cos 2y + 1} = \frac{e^{4x} - 1}{e^{4x} - 2e^{2x} \cos 2y + 1} - \frac{2e^{2x} \sin 2y}{e^{4x} - 2e^{2x} \cos 2y + 1}$$

$$W = U + iV$$

при  $x=c$   $\begin{cases} U = \frac{e^{4c} - 1}{e^{4c} - 2e^{2c} \cos 2y + 1} = \frac{e^{4c} - 1}{e^{4c} - 2e^{2c} \cos 2y + 1} \\ V = \frac{-2e^{2c} \sin 2y}{e^{4c} - 2e^{2c} \cos 2y + 1} = \frac{-2e^{2c} \sin 2y}{e^{4c} - 2e^{2c} \cos 2y + 1} \end{cases}$

$$e^{4c} - 2e^{2c} \cos 2y + 1 = \frac{e^{4c} - 1}{U}$$

$$V = \frac{-2e^{2c} \sin 2y U}{e^{4c} - 1}$$

$$\sin 2y = \frac{V(e^{4c} - 1)}{-2e^{2c} U}$$

$$y = \arcsin \frac{V(e^{4c} - 1)}{-2e^{2c} U}$$



$$u = \frac{e^{4c} - 1}{e^{4c} - 2e^{2c} \cos(\arcsin \frac{u(e^{4c} - 1)}{-2e^{2c}u})} + 1$$

$$u = \frac{e^{4c} - 1}{e^{4c} - 2e^{2c} \sqrt{1 - \frac{v^2(e^{4c} - 1)^2}{4e^{4c}u^2}}} + 1$$

$$u \left( e^{4c} - \frac{2e^{2c}}{2e^{2c}} \frac{\sqrt{4e^{4c}u^2 - v^2(e^{4c} - 1)^2}}{u} + 1 \right) = e^{4c} - 1$$

$$u(e^{4c} - \frac{\sqrt{4e^{4c}u^2 - v^2(e^{4c} - 1)^2}}{u} + 1) = e^{4c} - 1$$

$$ue^{4c} + u - e^{4c} + 1 = \sqrt{4e^{4c}u^2 - v^2(e^{4c} - 1)^2}$$

$$(ue^{4c} + 1) + (1 - e^{4c}) = \sqrt{4e^{4c}u^2 - v^2(e^{4c} - 1)^2}$$

$$u^2((e^{4c} + 1)^2 - 4e^{4c}) + 2u(e^{4c} + 1)(1 - e^{4c}) + (1 - e^{4c})^2 + v^2(e^{4c} - 1)^2 = 0$$

$$\frac{u^2((e^{4c} + 1)^2 - 4e^{4c})}{(e^{4c} - 1)^2} + \frac{2u(e^{4c} + 1)(1 - e^{4c})}{(e^{4c} - 1)^2} + \frac{(1 - e^{4c})^2}{(e^{4c} - 1)^2} + v^2 = 0$$

$$u = \frac{e^{4c} + 1}{e^{4c} - 1} = \cosh 2c$$

$$u^2(a^2 - a^2 + 1) - 2ua - \frac{(e^{4c} - 1)(1 - e^{4c})}{(e^{4c} - 1)(e^{4c} - 1)} + v^2 = 0$$

$$u^2 - 2ua + v^2 = -1$$

$$u^2 - 2ua + a^2 + v^2 = -1 + a^2$$

$$(u - a)^2 + v^2 = a^2 - 1$$

Получили  $v^2 + (u - a)^2 = a^2 - 1$  — уравнение окружности радиуса  $|a| > 1$ ,  
 $a = \cosh 2c$ , очевидным образом  $u = \pm 1$



$$\text{npu } y=C \begin{cases} u = \frac{e^{4x}-1}{e^{4x}-2e^{2x}\cos 2C+1} \\ v = \frac{-2e^{2x}\sin 2C}{e^{4x}-2e^{2x}\cos 2C+1} \end{cases}$$

$$ue^{4x} - 2ue^{2x}\cos 2C + u = e^{4x} - 1$$

$$ue^{4x} - e^{4x} = 2ue^{2x}\cos 2C - u - 1$$

$$e^{4x}(u-1) = 2ue^{2x}\cos 2C - u - 1$$

$$e^{4x} = \frac{2ue^{2x}\cos 2C - u - 1}{u-1}$$

$$u = \frac{-2e^{2x}\sin 2C}{\frac{2ue^{2x}\cos 2C - u - 1}{u-1} - 2e^{2x}\cos 2C + 1}$$

$$v = \frac{-2e^{2x}\sin 2C(u-1)}{2ue^{2x}\cos 2C - u - 1 - (2e^{2x}\cos 2C - 1)(u-1)}$$

$$v(2ue^{2x}\cos 2C - u - 1 - 2e^{2x}\cos 2C u + 2e^{2x}\cos 2C + u - 1) = -2e^{2x}\sin 2C(u-1)$$

$$v(-2 + 2e^{2x}\cos 2C) = -2e^{2x}\sin 2C(u-1)$$

$$-2v + 2ve^{2x}\cos 2C = -2e^{2x}\sin 2C(u-1) \quad | : -2$$

$$v - ve^{2x}\cos 2C = e^{2x}\sin 2C(u-1)$$

$$v = e^{2x}\sin 2C(u-1) + ve^{2x}\cos 2C$$

$$e^{2x} = \frac{v}{\sin 2C(u-1) + v\cos 2C} \Rightarrow 2x = \ln \frac{v}{\sin 2C(u-1) + v\cos 2C}$$



$$X = \ln \left( \frac{U}{\frac{\sin 2C(u-1) + V \cos 2C}{2}} \right)$$

$$u = \frac{e^{2 \ln \left( \frac{U}{\sin 2C(u-1) + V \cos 2C} \right)} - 1}{e^{2 \ln \left( \frac{U}{\sin 2C(u-1) + V \cos 2C} \right)} - 2e^{\ln \left( \frac{U}{\sin 2C(u-1) + V \cos 2C} \right)} \cos 2C + 1}$$

$$u = \frac{\left( \frac{U}{\sin 2C(u-1) + V \cos 2C} \right)^2 - 1}{\left( \frac{U}{\sin 2C(u-1) + V \cos 2C} \right)^2 - 2 \left( \frac{U}{\sin 2C(u-1) + V \cos 2C} \right) \cos 2C + 1}$$



$$U^2 U - u U^2 \cos^2 2C + u(u-1)^2 \sin^2 2C - U^2 + \sin^2 2C (u-1)^2 + \sin 4C \cdot U(u-1) + \frac{U^2 \cos^2 2C}{u-1} = 0$$

$$U^2 (u - u \cos^2 2C - 1 + \cos^2 2C) + (u-1)^2 (u \sin^2 2C + \sin^2 2C + \sin 4C \frac{U}{u-1}) = 0$$

$$U^2 u (1 - \cos^2 2C) + U^2 (-1 + \cos^2 2C) + (u-1)^2 ((u+1) \sin^2 2C + 2 \sin 2C \cos 2C \frac{U}{u-1}) = 0$$

$$U^2 u \sin^2 2C + U^2 \sin^2 2C + (u-1)^2 \sin 2C ((u+1) \sin 2C + 2 \cos 2C \frac{U}{u-1}) = 0$$

$$\cancel{\sin^2 2C \cos^2 2C} + U^2 \cancel{\sin^2 2C} (u-1) + (u-1)^2 \sin^2 2C (u+1) + 2 \cos 2C U (u-1) \sin 2C = 0$$

$$U^2 + \frac{(u-1)^2 \sin^2 2C (u+1)}{\sin^2 2C (u-1)} + \frac{2 \cos 2C U (u-1) \sin 2C}{\sin^2 2C (u-1)} = 0$$

$$U^2 + (u-1)(u+1) + 2 \cot 2C \frac{U}{\sin 2C} = 0 \quad | \cdot \sin 2C$$

$$U^2 + u^2 - 1 + 2 \cot 2C \frac{U}{\sin 2C} = 0$$

$$U^2 + 2 \cot 2C U + u^2 = 1$$

$$U^2 + 2 \cot 2C U + \cot^2 2C + u^2 = 1 + \cot^2 2C$$

$$(U + \cot 2C)^2 + u^2 = 1 + \cot^2 2C$$

$$u^2 + (U + 6)^2 = 1 + 6^2 \quad (6 = \cot 2C) -$$

Получили уравнение окружности в плоскости  $u = \pm 1$



$$3) W = e^{z^2} = e^{(x+yi)^2} = e^{x^2+2xyi-y^2} = e^{x^2-y^2} \cdot e^{2xyi} =$$

$$= e^{x^2-y^2} (\cos 2xy + i \sin 2xy) = u + iv$$

$$u = e^{x^2-y^2} \cos 2xy$$

$$v = e^{x^2-y^2} \sin 2xy$$

now  $x=c$

$$u = e^{c^2-y^2} \cos 2cy$$

$$v = e^{c^2-y^2} \sin 2cy$$

$$u^2 + v^2 = e^{2(c^2-y^2)} (\cos^2 2cy + \sin^2 2cy)$$

$$u^2 + v^2 = e^{2(c^2-y^2)} = e^{2c^2-2y^2}$$

now  $y=c$

$$u = e^{x^2-c^2} \cos 2xc$$

$$v = e^{x^2-c^2} \sin 2xc$$

$$u^2 + v^2 = e^{2(x^2-c^2)}$$

$$2(x^2-c^2) = 2c^2 - 2y^2$$

$$2x^2 + 2y^2 = 2c^2 + 2c^2$$

$$2(x^2 + y^2) = 4c^2$$

Letting  $u = \rho \cos \phi$   $u^2 + v^2 = e^{2(x^2-c^2)}$  (now  $y=c$ )

$$v = \rho \sin \phi$$

$$\rho^2 = e^{2(x^2-c^2)}$$

$$\rho = e^{x^2-c^2}$$

$$u + iv = e^{x^2-c^2} (\cos 2cx + i \sin 2cx) = e^{x^2-c^2} \cdot e^{2cxi} = \rho (\cos \phi + i \sin \phi) = e^{x^2-c^2} e^{2cxi}$$



Решение:

$$\begin{cases} x^2 - c^2 = \ln p \\ 2cx = p \end{cases} \Rightarrow \begin{cases} p = e^{x^2 - c^2} = \sqrt{u^2 + v^2} \Rightarrow p^2 = e^{2(x^2 - c^2)} = u^2 + v^2 \\ x = \frac{p}{2c} \Rightarrow x^2 = \frac{p^2}{4c^2} \end{cases}$$

N1.2

при  $y = c$  получим единичный круг радиуса  $p = \frac{p^2}{4c^2} - c^2$

при  $y = 0$   $p = e^{x^2} = \sqrt{u^2} \Rightarrow p^2 = e^{2x^2}$

$$u + v_i = e^{x^2 - c^2} (\cos^2 2\varphi - \sin^2 2\varphi) = e^{x^2} = e^{\frac{p^2}{4c^2}}$$

$$p^2 = e^{2x^2} = (p \cos \varphi)^2 + (p \sin \varphi)^2 = e^{2(x^2 - c^2)} (\cos^2 2\varphi + \sin^2 2\varphi) = e^{2(x^2 - c^2)}$$

$$x^2 = \ln p$$

$$\ln p = \frac{p^2}{4c^2}$$

$$4c^2 \ln p = p^2$$

при  $1 \leq p < \infty$   $\ln p$  - монотонно  $p = 0$  (при  $c = 0$ )

Решая: при  $y = 0$  (или) наоборот. при  $0 = 0, 1 \leq p < \infty$

$$\text{При } x = c \quad \begin{cases} c^2 - y^2 = \ln p \\ 2cy = p \end{cases} \Rightarrow \begin{cases} p = e^{c^2 - y^2} \Rightarrow p^2 = e^{2(c^2 - y^2)} = u^2 + v^2 \\ y^2 = \frac{p^2}{4c^2} \end{cases}$$

при  $x = c$  получим  $p = e^{c^2 - \frac{p^2}{4c^2}}$  - единичный круг радиуса

при  $x = 0$   $\begin{cases} -y^2 = \ln p \\ p = 0 \end{cases} \Rightarrow$  рассмотрим при  $0 < p \leq 1$  - получим окружность  $p = 0$  (или ось  $x = 0$ )